

PORA ACADEMY CAPSTONE PROJECT GUIDE

Project Title

Setup, Implementation, Configuration, and Validation of Snort/Suricata Intrusion Detection System (IDS)

1. Project Overview

The purpose of this capstone project is to assess students' ability to deploy, configure, and manage a Network Intrusion Detection System (NIDS) using either Snort or Suricata.

Students will work collaboratively in groups to design, implement, monitor, and validate an IDS solution within a controlled lab environment. The project simulates a real-world Security Operations Center (SOC) deployment where security analysts must monitor network traffic, detect malicious activities, and create custom detection rules.

2. Learning Objectives

At the completion of this project, students should be able to:

- Understand the architecture and operation of IDS technologies.
- Deploy and configure Snort or Suricata.
- Monitor network traffic and analyze alerts.
- Create and test custom IDS rules.
- Generate malicious and benign traffic for validation.
- Investigate IDS detections.
- Produce professional technical documentation.
- Present findings and recommendations.

3. Group Requirements

Team Composition

Students must work in groups of 3 or 4

Each group should assign roles such as:

- Project Lead
- IDS Engineer
- Threat Simulation Analyst
- Documentation & Reporting Lead

Note: One student may perform multiple roles depending on group size.

4. Project Scenario

Your organization requires a Network Intrusion Detection System (IDS) to monitor internal network traffic and identify suspicious activities.

Your team has been tasked with:

1. Deploying Snort or Suricata.
2. Configuring the IDS.
3. Creating custom detection rules.
4. Testing and validating detections.
5. Producing a professional implementation report.

5. Project Tasks

Task 1 – Lab Design

Create a lab environment containing:

Minimum Components

- 1 IDS Server (Snort or Suricata)
- 1 Attacker Machine
- 1 Victim Machine

Provide:

- Network Diagram
- System Specifications

Task 2 – IDS Installation

Install either:

- Snort
- OR
- Suricata

Document:

- Installation steps
- Packages installed
- Configuration files modified
- Challenges encountered
- Screenshots

Task 3 – IDS Configuration

Configure the IDS to:

- Monitor the appropriate network interface
- Load custom rule files
- Generate alerts
- Log events

Document:

- Configuration changes
- Network interface selected
- Logging settings
- Alert output settings

Include screenshots.

Task 4 – Custom Rule Development

Each group must develop and implement a minimum of:

Six (6) Custom Rules

The rules should detect different activities.

Examples:

1. ICMP Ping Detection
2. SSH Connection Detection
3. HTTP GET Request Detection
4. Nmap Scan Detection
5. FTP Login Attempt Detection
6. Custom Malware Test String Detection

Each rule must include:

- Rule Objective
- Rule Syntax
- Explanation of Rule Components
- Expected Alert Output

Task 5 – Traffic Generation & Validation

Generate traffic to test all rules.

Examples:

- Ping traffic
- Nmap scans
- HTTP requests
- FTP connections
- SSH sessions
- Custom payload traffic

For each rule:

Validation Requirements

Provide:

- Test Procedure

- Command Used
- Traffic Generated
- Alert Generated
- Screenshot Evidence

Demonstrate that each rule successfully triggers an alert.

Task 6 – Analysis of Alerts

Analyze generated alerts and explain:

- Source IP
- Destination IP
- Timestamp
- Protocol
- Triggered Rule
- Security Relevance

Provide screenshots and observations.

6. Deliverables

Each group must submit:

1. GitHub Repository

The repository should contain:

Documentation

- README.md
- Project Overview
- Lab Setup
- Installation Guide
- Configuration Guide
- Testing Procedures
- Validation Results
- Lessons Learned

Configuration Files

- snort.conf or suricata.yaml
- Custom Rules File

Evidence

- Screenshots
- Network Diagram

2. Professional Technical Report

Format:

- Microsoft Word (.docx)
- PDF (Optional)

Recommended Length:

15–25 Pages

The report must contain:

- Executive Summary
- Introduction
- Objectives
- Lab Design
- Installation Process
- Configuration Process
- Custom Rules
- Validation & Testing
- Results
- Challenges Encountered
- Recommendations
- Conclusion
- References

7. Evaluation Criteria

Criteria	Weight
Lab Design & Architecture	10%
IDS Installation	15%
IDS Configuration	15%
Quality of Custom Rules	20%
Validation & Testing	20%
Documentation Quality	10%
GitHub Repository Quality	5%
Presentation & Professionalism	5%

8. Bonus Opportunities (+10%)

Additional marks may be awarded for:

- More than six custom rules
- Integration with SIEM platforms
- Advanced attack simulations
- Dashboard creation
- Automation scripts
- Threat intelligence integration

Maximum Bonus: 10%

9. Submission Guidelines

Submission Package:

- GitHub Repository Link

- Technical Report (Word Document)
- Presentation Slides

Naming Convention:

GroupNumber_ProjectName

Example:

Group03_Suricata_IDS_Implementation

Submission Deadline: